RFI Report for September 2024

Executive summary

The few samples selected in the month of September show that the cross correlation products are clean overall, whereas the autocorrelation is affected by relatively weak RFI. Nevertheless, the Laduma observation on the 8th of September appears to have been contaminated by prominent RFI emission whose minimum strength is about 0.4 and the maximum one reaches 0.6 (see subsection a). These contaminants are known to be from Vodacom and MTN downlinks according to the investigation conducted by the RFI team, and the sub-bands they are seen at are not supposed to be used around site.

Overall, for the month of September, the RFI levels are relatively stable, with an overall average of 10%. Figure 15 indicates that the observed RFI primarily originates from a well-known source at UHF, specifically the GSM downlink frequencies

1. Introduction

The purpose of this document is to report the current status of RFI on-site. Given the construction work that has been going on over the last couple of months, it is therefore imperative to monitor the construction related activities on site that could potentially generate RFI signals which, in turn, contaminate the science observations. Preferably these reports are to be released on a weekly basis (or every two weeks).

2. Method

The flags data, specifically *ingest_flags*, are considered in these analyses, as they can be used as proxies for RFI frequency occupancy, i.e. RFI emission as a function of frequency channels. The strategy is to select two to three observations per week and report the status of the RFI contamination on site based on the flags of each dataset (of each observation). For each observation

- Autocorrelation flags of a few *scans* corresponding to the *gain calibrator* are selected. This is due to the fact that, at the time of writing, there are some issues with accessing a big chunk of an entire dataset.
- Autocorrelation flags, which are three dimensional arrays, i.e. *time*: *frequency channels*: *correlation products*, are averaged over both time and correlation products.
- Autocorrelation flags become a one dimensional array which is plotted in order to check the RFI occupancy in the frequency channels.

The cross correlation flags are also investigated in order to see how much the RFI emission affects the correlation products which are the main interest in an interferometric observation. The procedure for the cross correlation flags is the same as that of autocorrelation ones. To further go into more detail, the flags corresponding to short and long baselines are split into two parts, and each part is plotted separately. Based on some datasets that were analyzed

previously, the U-band observations are the ones that are most affected by the current RFI emissions on site. Hence, the majority of the datasets that will be inspected is in U-band, nevertheless observations in both L and S bands will be looked at too.
In what follows, it is worth noting that the vertical gray shaded areas in all plots denote the known RFI emissions in U-band.

3. Results

a. 2024, 08 September

The autocorrelation, short and long baseline cross correlation flags of the dataset collected on the 8th of September 2024 are plotted in Figure 1, 2, and 3 respectively. What is striking is the strong interference in all correlation products, and it appears that the emissions were external, as implied by the fact that they were cross correlated. Results also show that the weaker RFI are mainly present in the autocorrelation products. An investigation carried out by the RFI team identifies the emissions at 768 - 778 MHz and 801 - 811 MHz to be Vodacom and MTN downlinks respectively, and they are not supposed to be used on site (based on information obtained from RFI team)

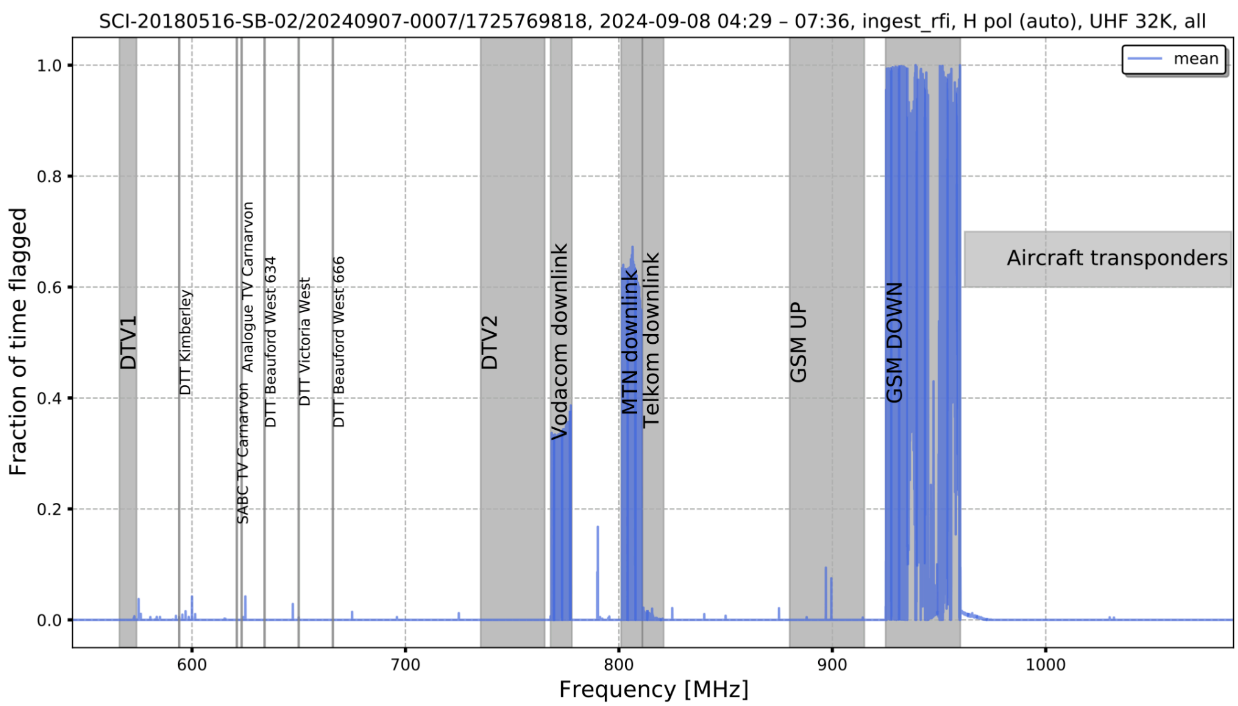

Figure 1: Mean of the autocorrelation flags of SB_ID:20240907-0007 dataset.

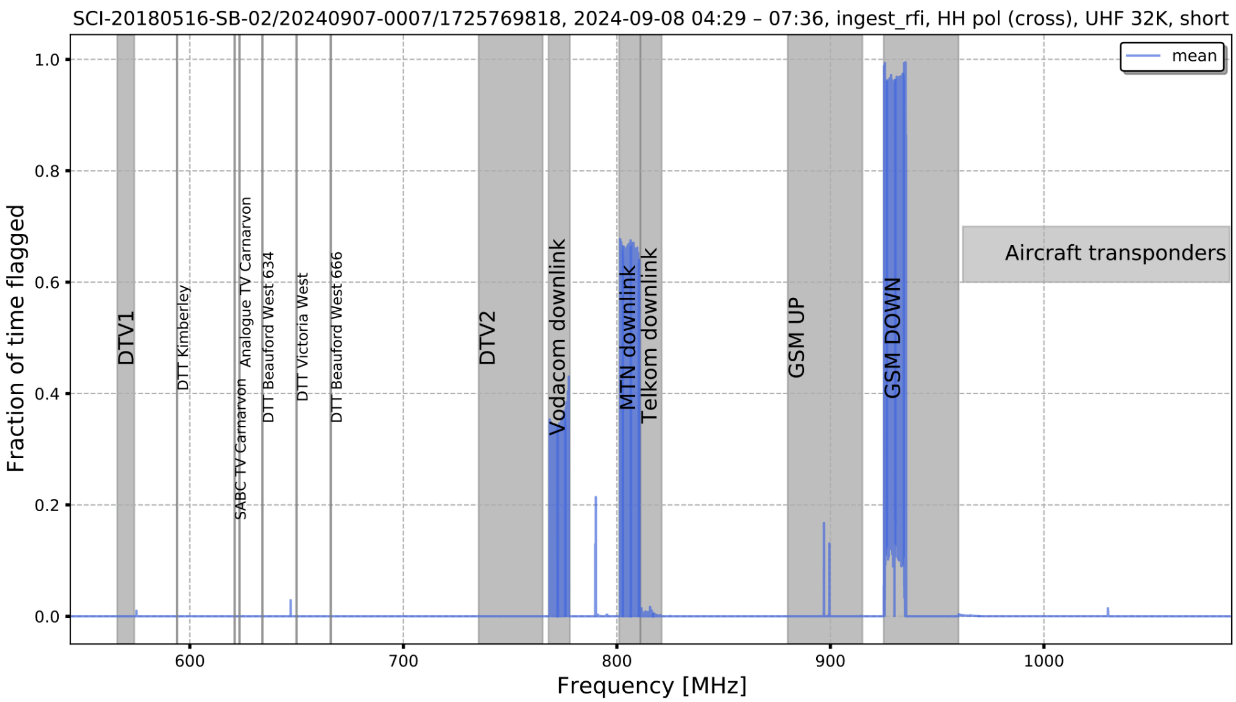

Figure 2: Mean of the cross correlation flags of the short baselines of SB_ID:20240907-0007 dataset.

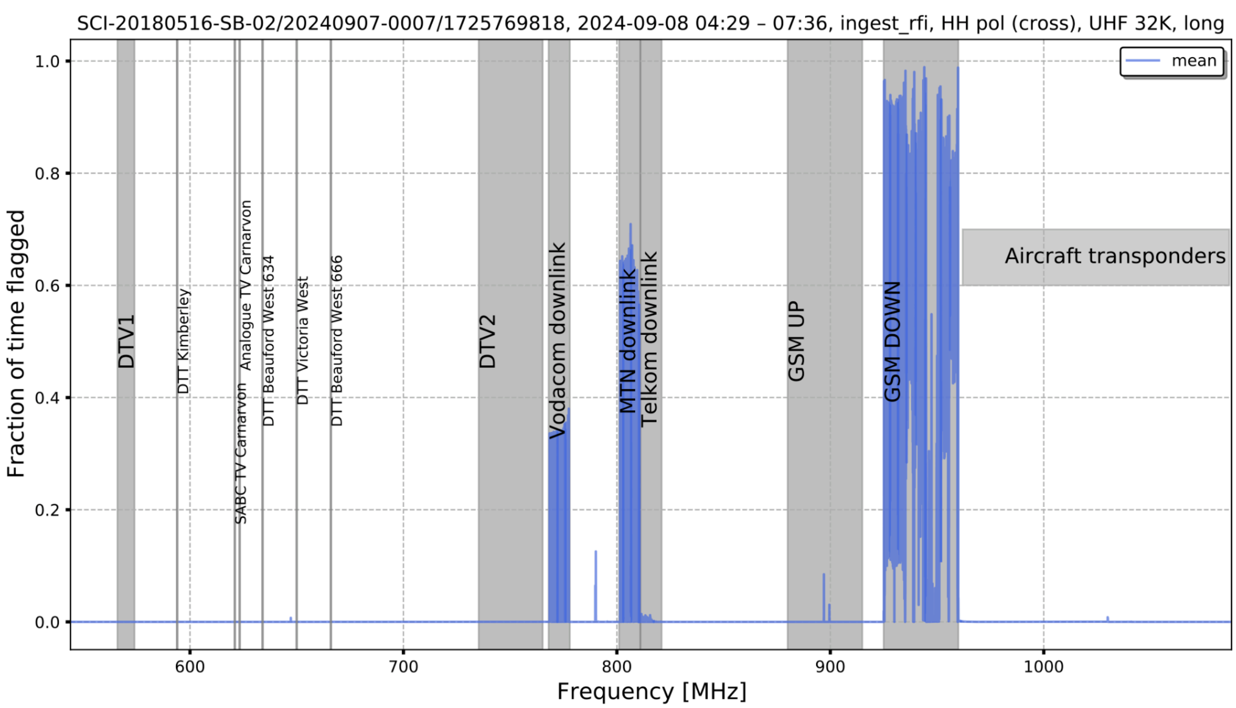

Figure 3: Mean of the cross correlation flags of long baselines of SB_ID:20240907-0007 dataset.

b. 2024, 13 September

The data observed on the 13th of September 2024 exhibit cross correlation products, as shown in Figures 5 and 6, that are less contaminated than the autocorrelation counterpart, presented in Figure 4 which contains unknown, but weak, RFI.

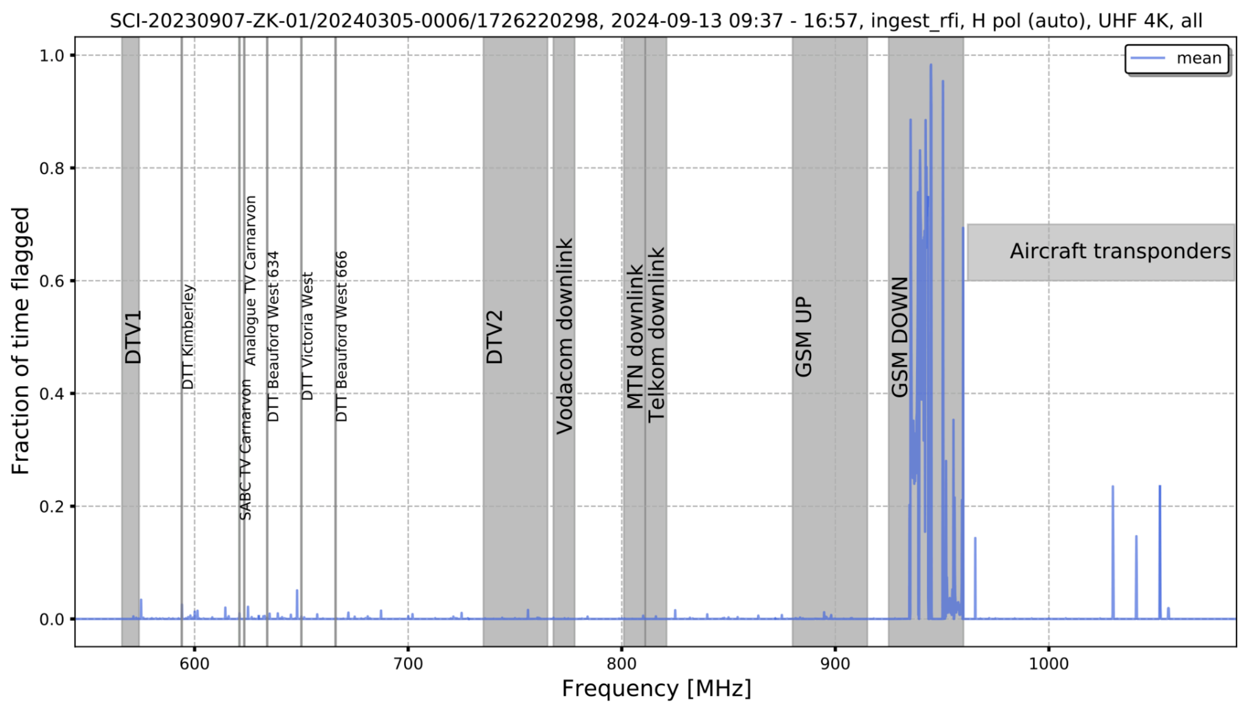

Figure 4: Mean of the autocorrelation flags of SB_ID:20240305-0006 dataset.

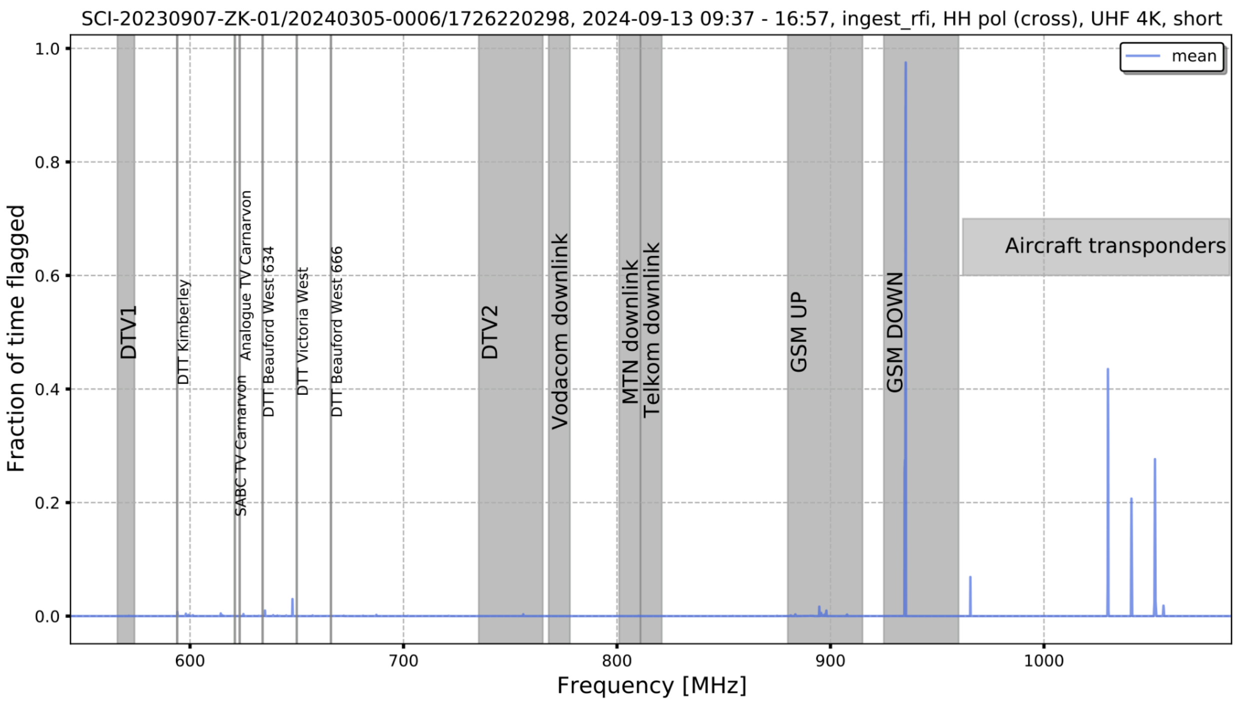

Figure 5: Mean of the cross correlation flags of short baselines of SB_ID:20240305-0006 dataset.

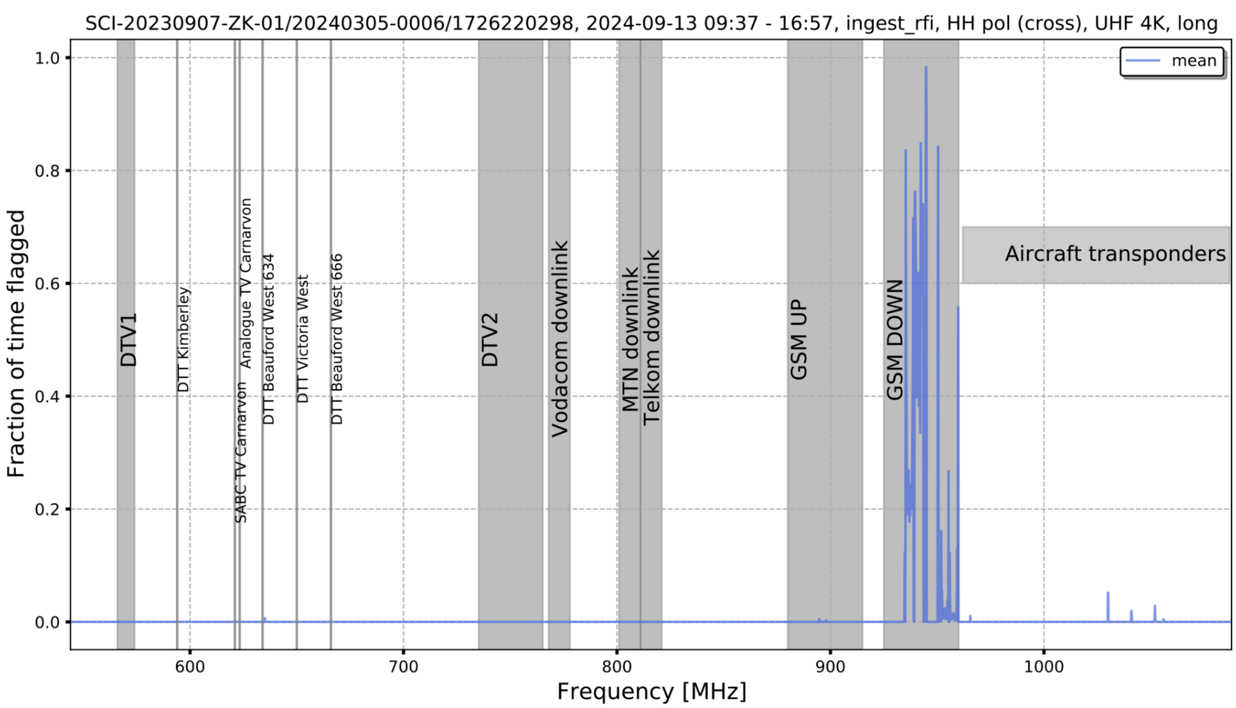

Figure 6: Mean of the cross correlation flags of long baselines of SB_ID:20240305-0006 dataset.

c. 2024, 18 September

The data that is considered in this section was collected on 18th of September 2024. Figures 7, 8 and 9 present the autocorrelation, short and long baseline cross correlation flags respectively. All the correlation products appear to be relatively clean, overall.

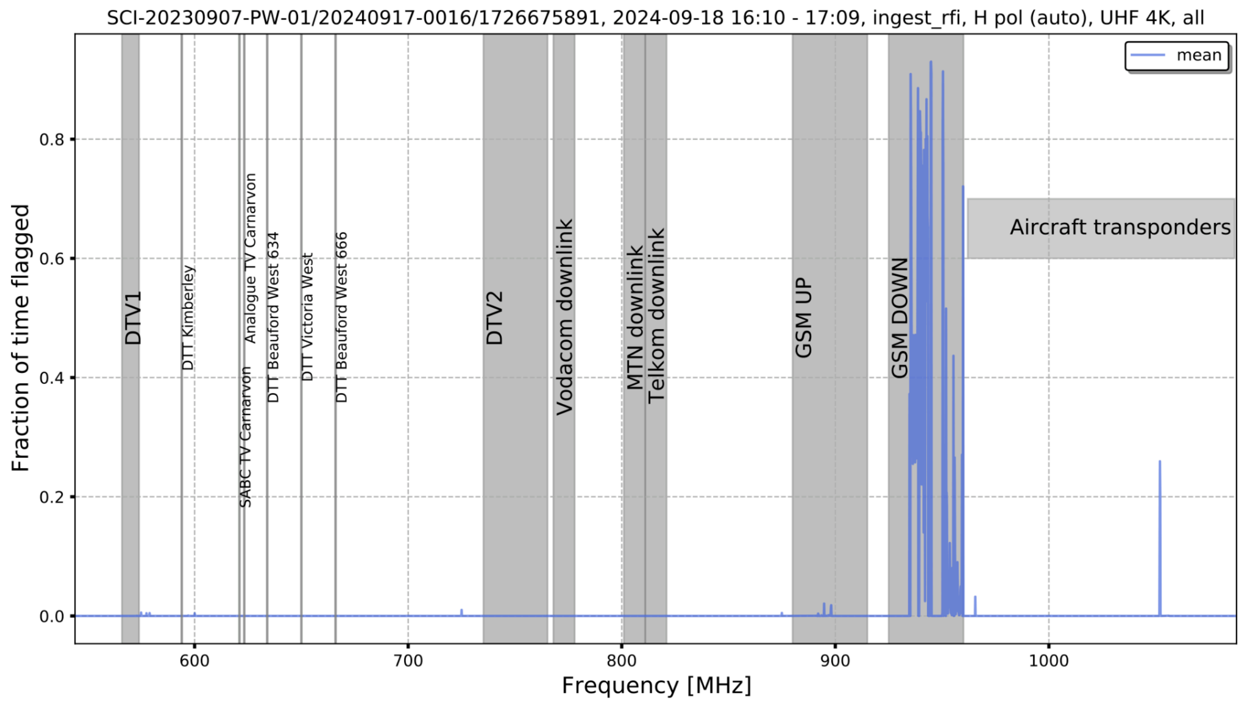

Figure 7: Mean of the autocorrelation flags of SB_ID:20240917-0016 dataset.

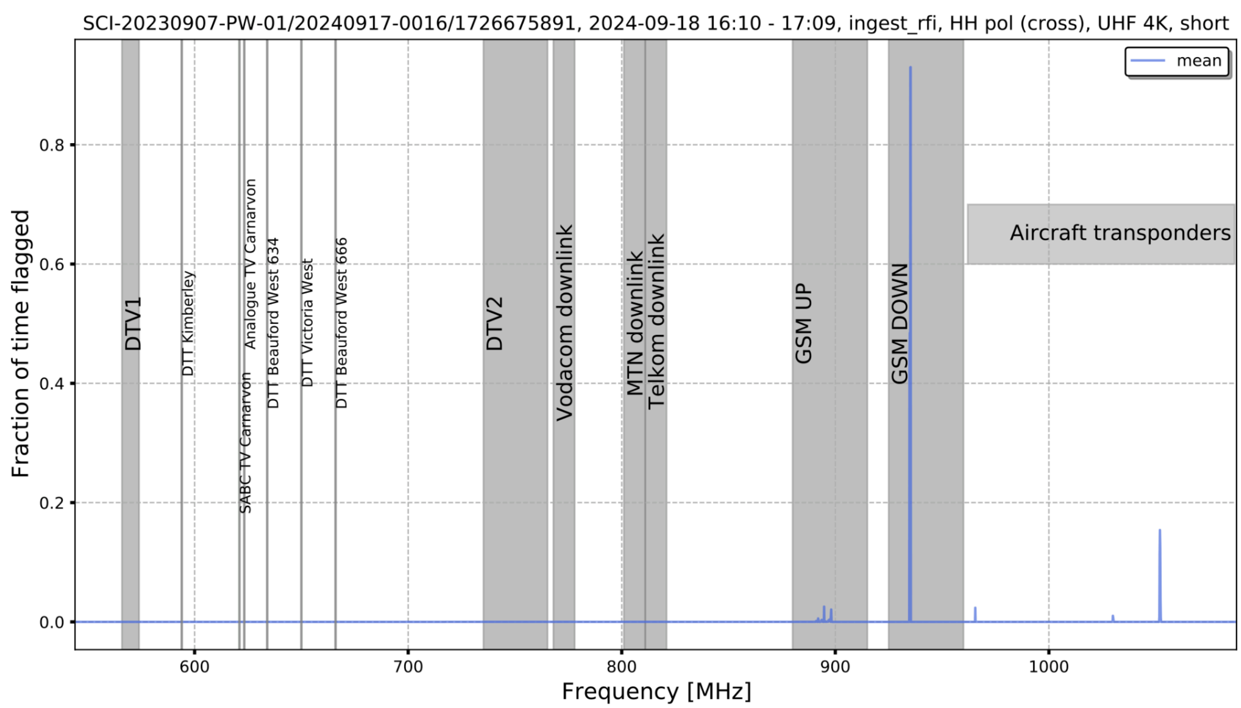

Figure 8: Mean of the cross correlation flags of the short baselines of SB_ID:20240917-0016 dataset.

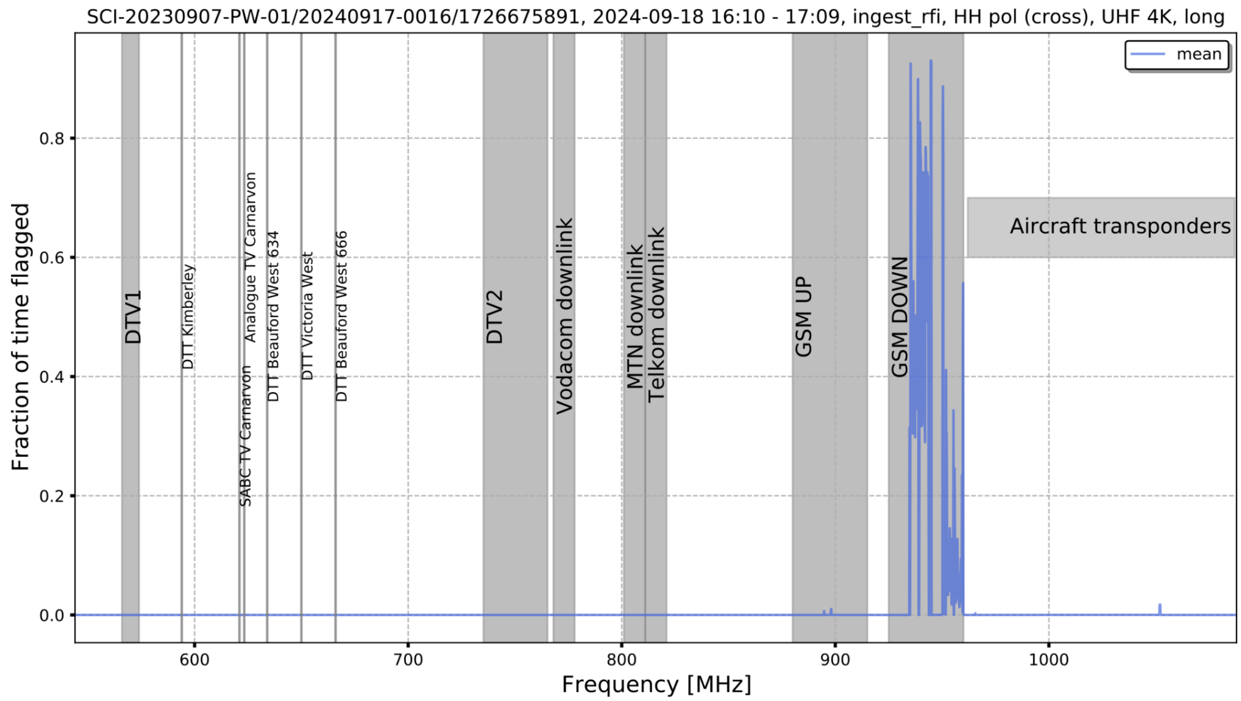

Figure 9: Mean of the cross correlation flags of long baselines of SB_ID:20240917-0016 dataset.

d. 2024, 27 September

The autocorrelation, obtained from a dataset collected on the 27th of September 2024, is contaminated by relatively weak interferences, as shown in Figure 10. Most of the small noises in Figure 10 are no longer noticeable in both short and long baseline cross correlation (see Figures 11 and 12).

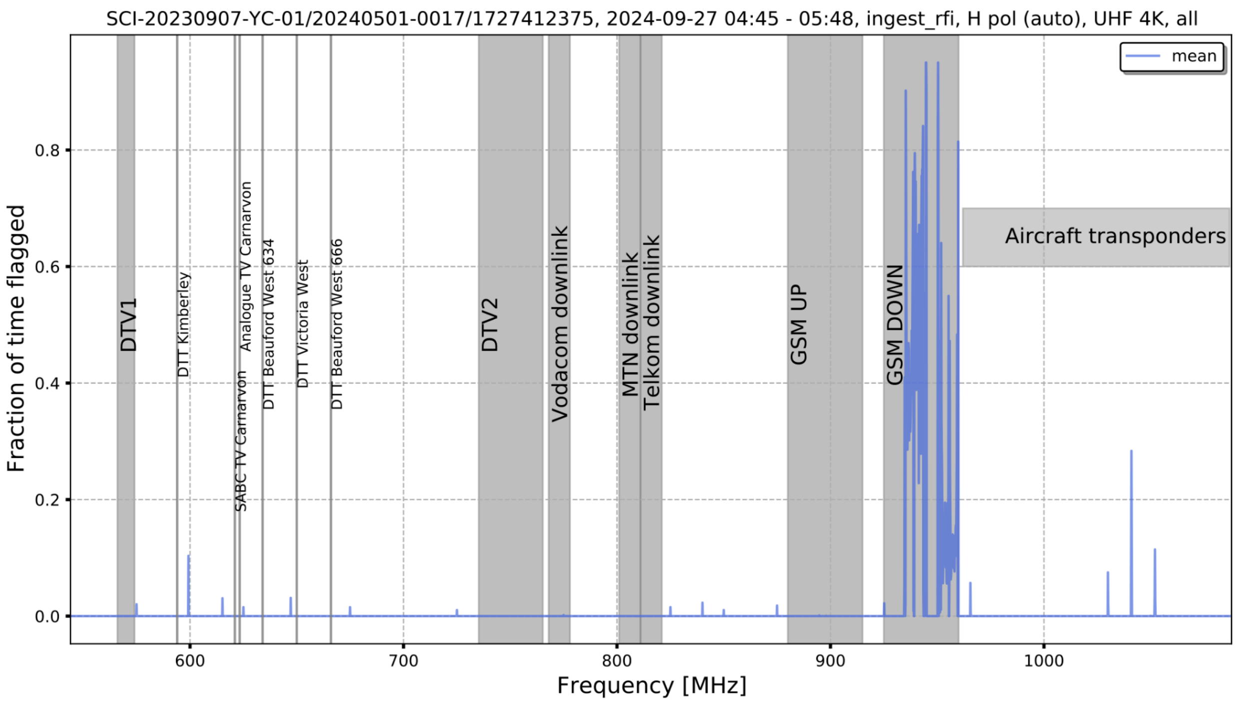

Figure 10: Mean of the autocorrelation flags of SB_ID:20240501-0017 dataset.

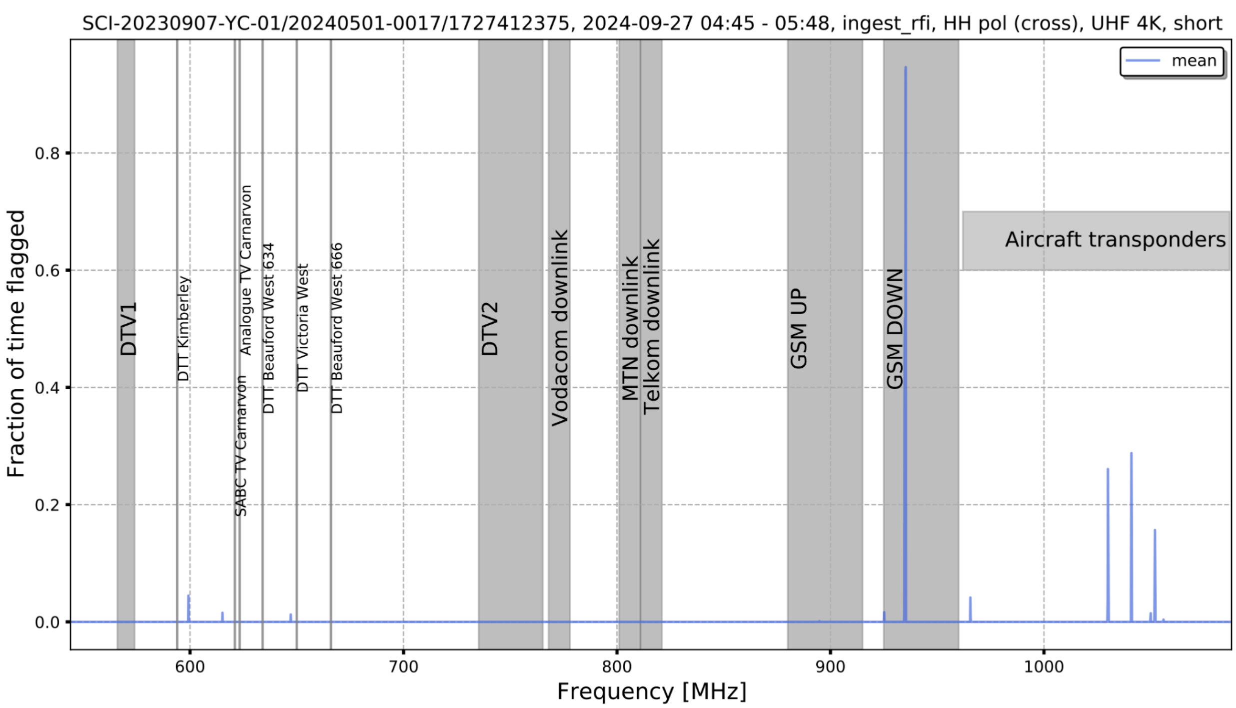

Figure 11: Mean of the cross correlation flags of short baseline of SB_ID:20240501-0017 dataset.

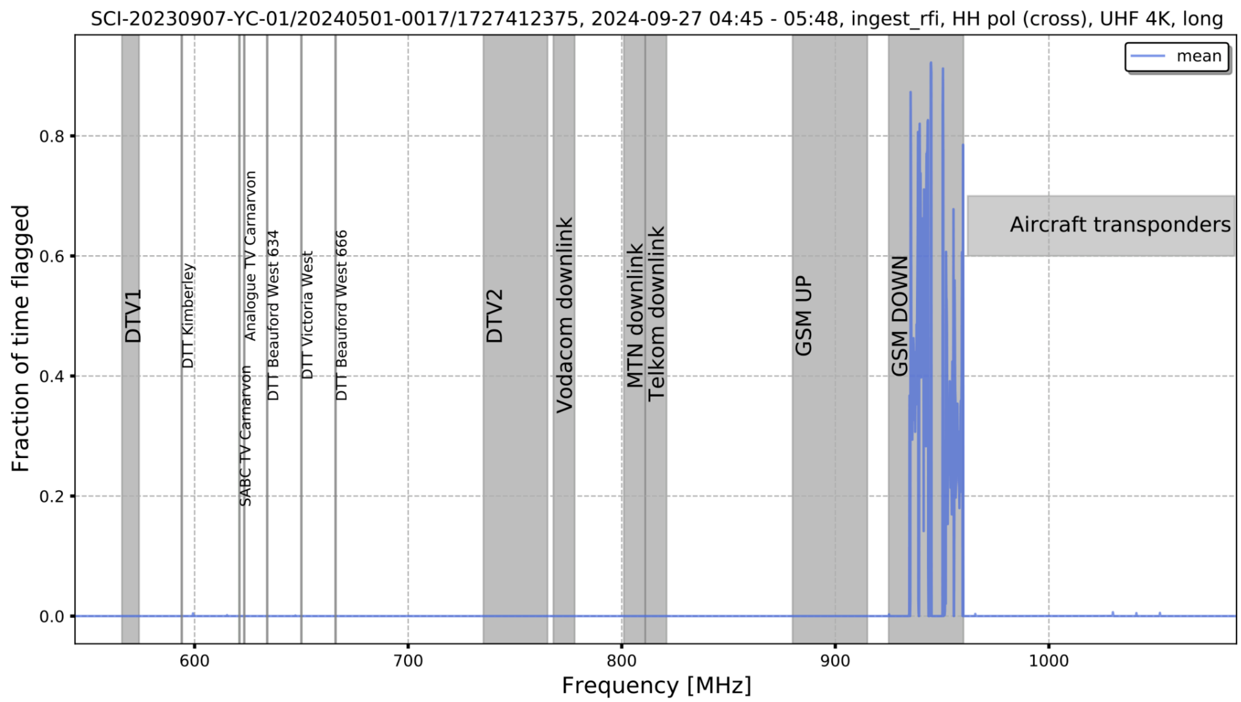

Figure 12: Mean of the cross correlation flags of long baselines of SB_ID:20240501-0017 dataset.

4 . RFI summary for August 2024 at UHF, HH cross polarization

Now we look at the overall RFI for the month of September by concatenating all science observations for the month. Here we select particularly 8s observations, observed at 4k or 32k channel mode. The monitoring and analysis of RFI are presented using the KATHPRFI pipeline [1], highlighting the impact of construction on RFI levels.

KATHPRFI algorithm

The kathprfi essentially processes two numpy arrays, Master and Counter, to update the number of RFI points and observations per voxel which refers to 5-D array of Time, Frequency, Baseline, Elevation and Azimuth. It iterates over chunks of frequency indices, baseline indices, and time indices. For each combination, the algorithm updates the Master array by the corresponding ‘Good flags‘ value and the Counter array by 1. Good flags for a given observation are based on a flag type, e.g. if flag type is calibration RFI then target tags for that observation will be obtained from a complete calibration report. The output of this is an xarray dataset that contains arrays of these two arrays and the dimensions of interest as mentioned above.

Computation of marginal probability

The RFI probability is computed by using two arrays from the output Zarr file obtained from the kathprfi pipeline : ’Master‘ and ‘Counter‘. These arrays represent the number of RFI points and the total number of observations, respectively. The function sums these arrays along a specified dimension to get total counts and then calculates the probability as the ratio of these sums.

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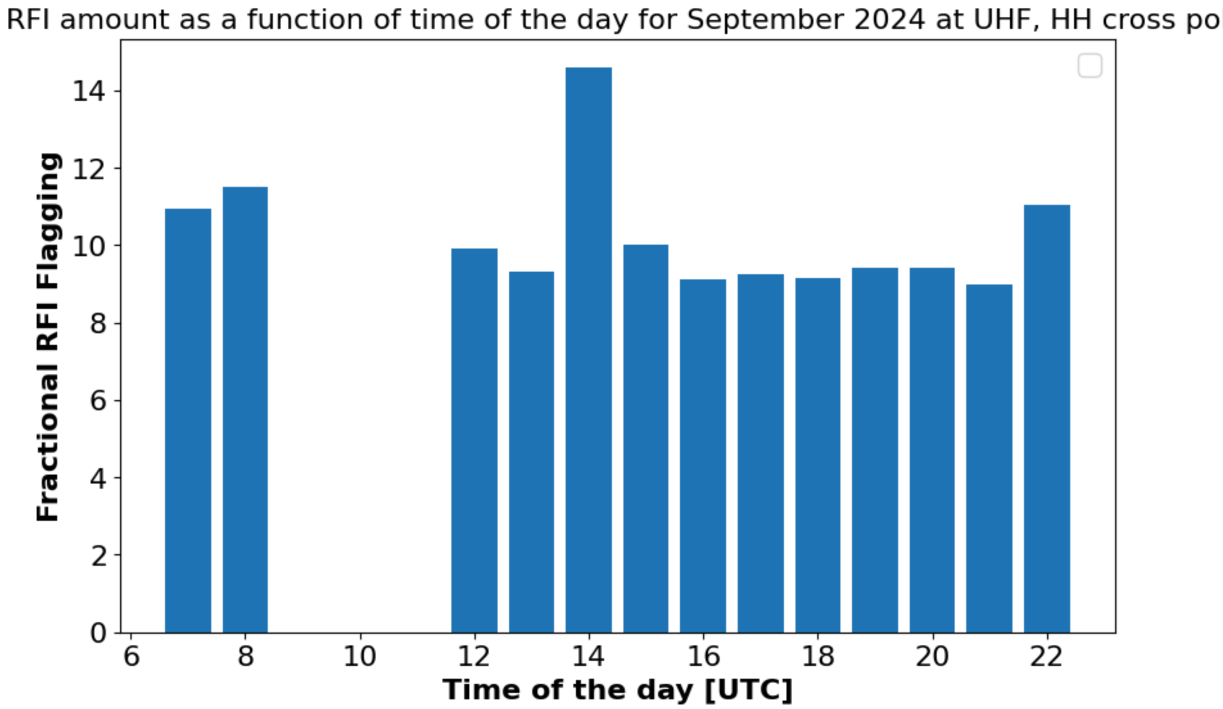

Figure 13: The plot above shows the fractional RFI flagging as a function of time of the day in UTC for August 2024 UHF horizontal cross pol observations.

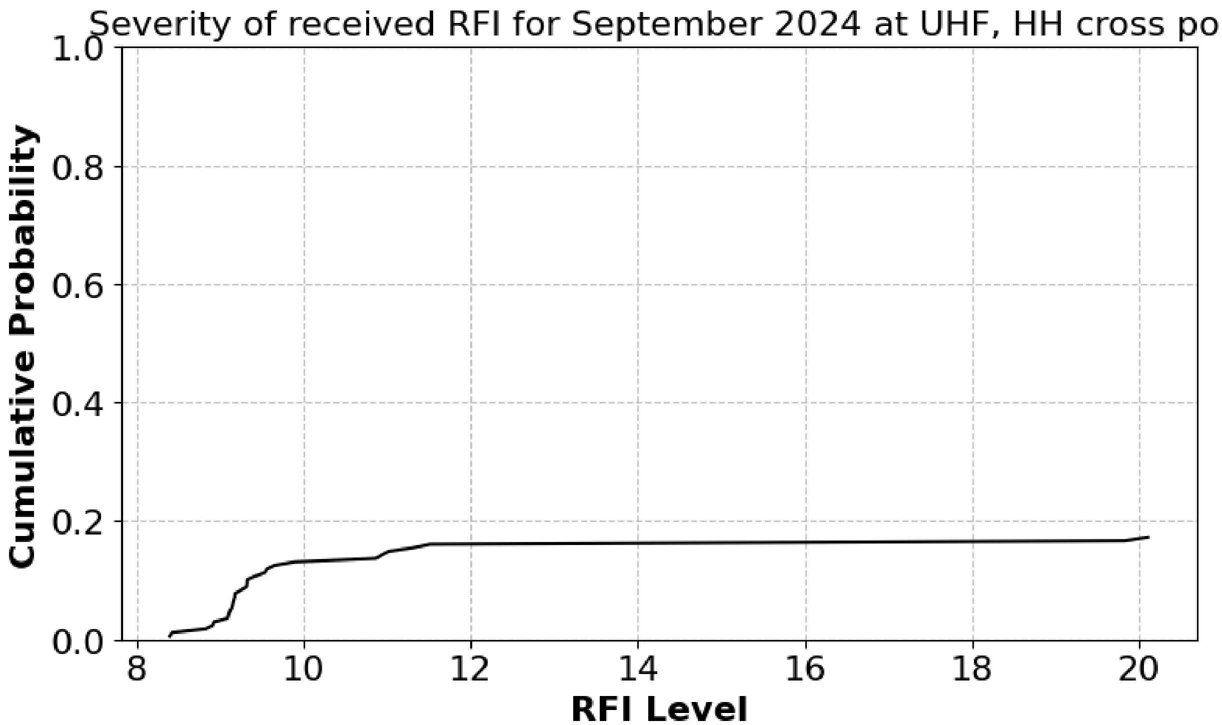

Figure 14: RFI distribution : cumulative probability of RFI as a function of RFI levels.

Figure 13 illustrates the hourly behavior of RFI for the month of August 2024. This view allows us to identify specific time periods affected by RFI. The probability of RFI is relatively low most of the time, with an average RFI percentage of 9%. At 14:00 UTC, the fractional RFI rises a bit to a maximum of 14%, this suggests a period when RFI is especially prevalent. This could be attributed to a number of causes: This could be due to specific construction activities on site , also could be due to a source under observation at the time. To be able to characterize, we need to look at the frequency spectra at that time of the time.

Figure 14, shows the Empirical Cumulative Distribution Function (ECDF) of RFI levels for August 2024. The x-axis represents the RFI levels, while the y-axis shows the cumulative probability, which represents the proportion of RFI data points that are less than or equal to each RFI level. This is useful for understanding the overall intensity distribution of the RFI events. For example, Figure 14 shows that between RFI level 8 and 10, around 16% of the data falls below level 10 with a steep slope from lowest probability, indicating that low RFI levels are common or very concentrated in that range. Therefore, we can conclude that around 16% of all RFI data points fall at or below an RFI level of 8. The steep initial rise in the ECDF indicates that a large portion of RFI events occur at these lower intensities, while higher RFI levels are much less frequent, occurring only in a small fraction of the observations. Thus, there are no severe RFI events for September.

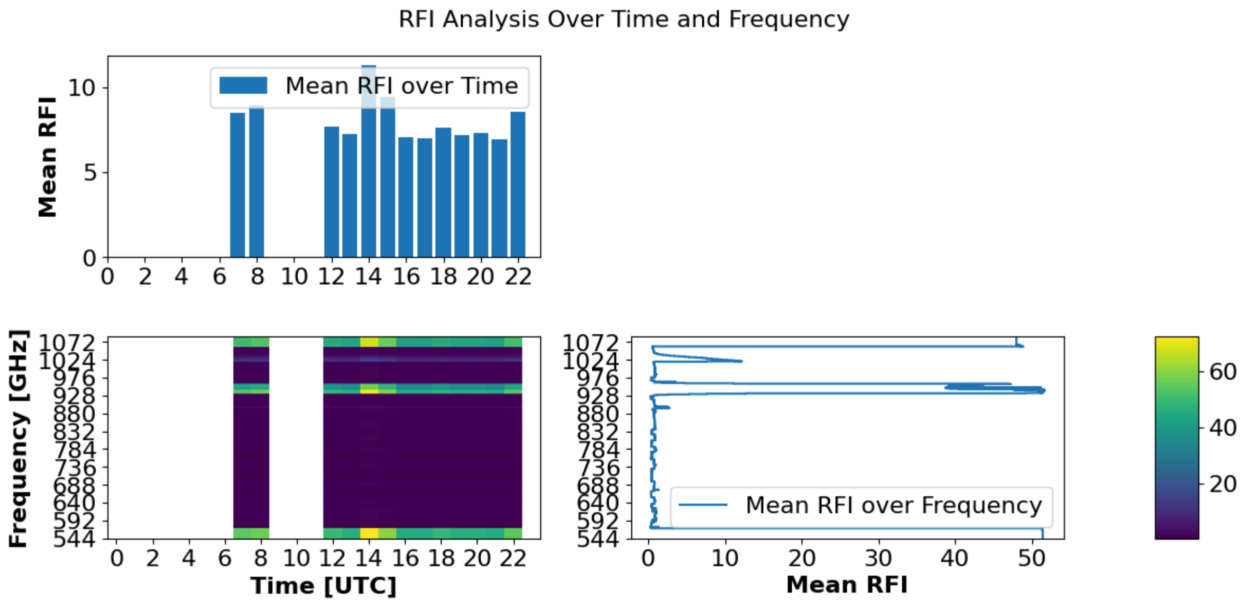

Figure 15 : Mean RFI across time and frequency space

Figure 15, depicts a 2 dimensional view of RFI patterns over time and frequency. The top 1D bar plot shows the mean RFI levels over time exactly the same as Figure 13. The 2D heatmap in the centre represents the detailed RFI activity across time and frequency, where white areas suggest missing data or non-monitored frequency intervals. Notably, intense RFI zones appear as bright yellow or green patches as indicated by the colour bar (far right), notably between 544-592, 800–900 MHz and at higher frequencies above 950 MHz, with strong RFI at 14:00 UTC of about 15%. The 1D plot on the right highlights mean RFI levels over different frequencies, with spikes indicating strong interference at specific frequencies. Fortunately we know the RFI present in these frequencies, GSM downlink – mobile communications. There is a weak spike around 784 MHz which is due to pilling.